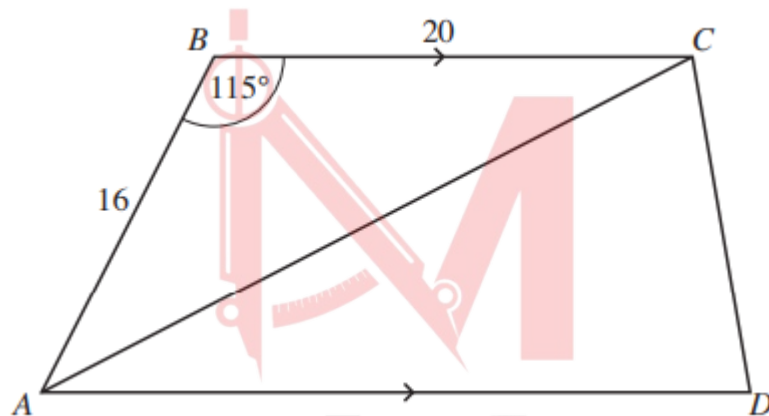


Trigonometry

Worksheet 2b

Further Trigonometry

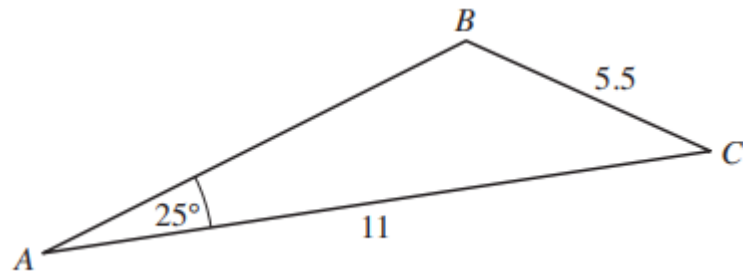
1. S11/qp21/q7



ABCD is a trapezium with AD parallel to BC. $AB = 16$ cm, $BC = 20$ cm and $\angle ABC = 115^\circ$.

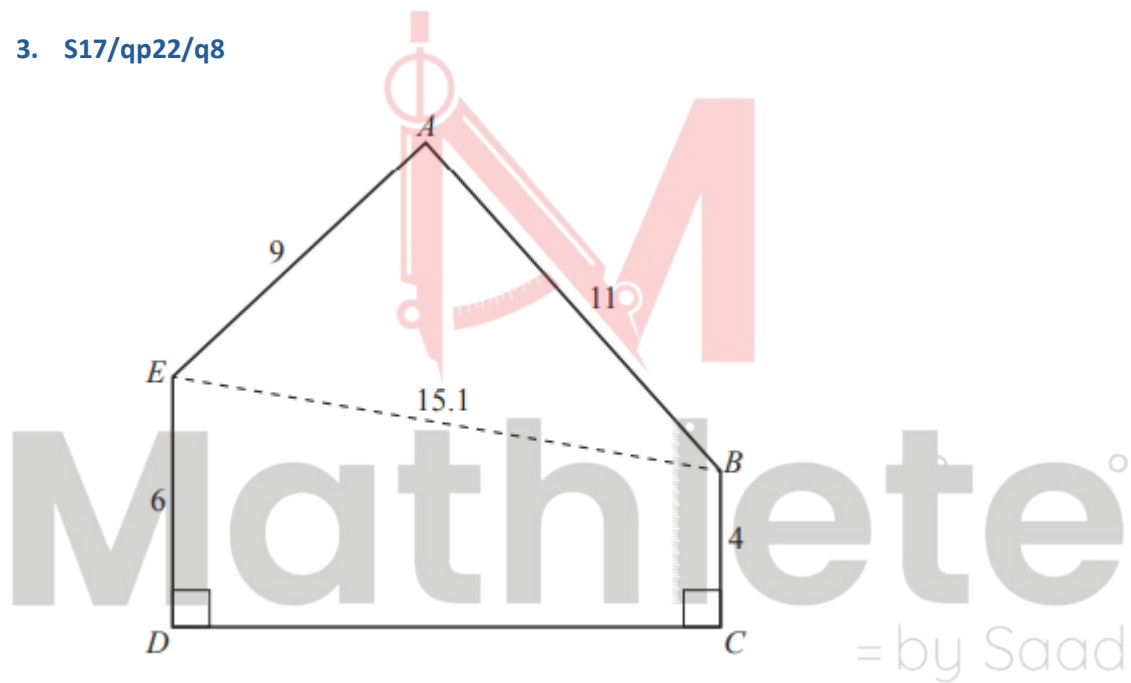
- a. Find AC. [4]
- b. Show that the perpendicular distance between BC and AD is 14.5 cm. [2]
- c. The area of the trapezium ABCD is 348 cm^2 .
 - i. Find AD. [2]
 - ii. Show that the area of triangle ACD is 203 cm^2 . [1]
 - iii. Hence, or otherwise, find $\angle CAD$. [3]

2. W11/qp21/q9(a)



In the diagram, $AC = 11$ cm, $BC = 5.5$ cm and $\angle BAC = 25^\circ$. It is given that $\angle ABC$ is an obtuse angle. Calculate $\angle ACB$. [4]

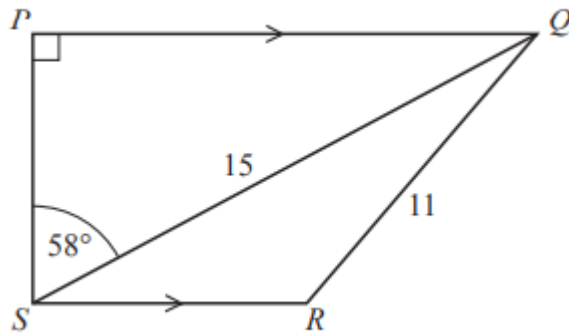
3. S17/qp22/q8



ABCDE is the cross-section of a building. All the lengths are given in metres.

- Calculate DC. [3]
- Calculate angle $\angle AEB$. [3]
- Calculate the area of the cross-section. [4]

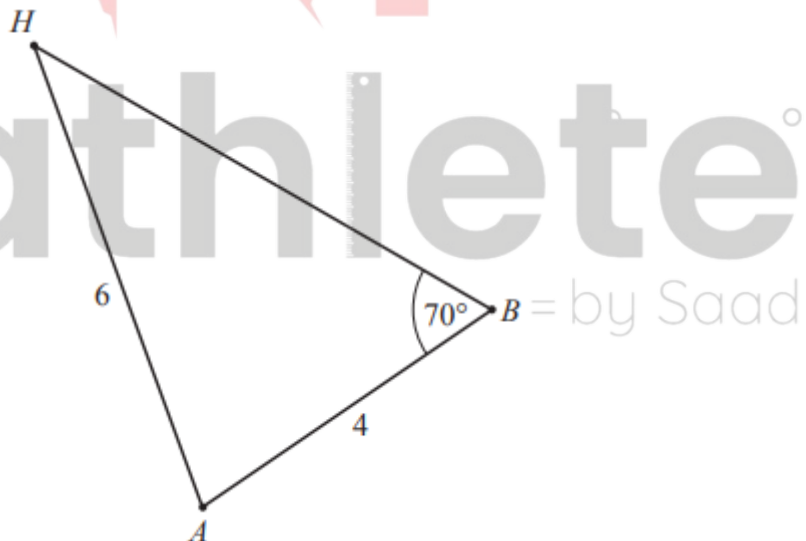
4. S19/qp21/q5(b)



PQRS is a trapezium with PQ parallel to SR and $\angle SPQ = 90^\circ$. $SQ = 15\text{cm}$, $QR = 11\text{cm}$ and $\angle PSQ = 58^\circ$.

- Calculate PS. [2]
- Calculate the obtuse angle $\angle SRQ$. [4]

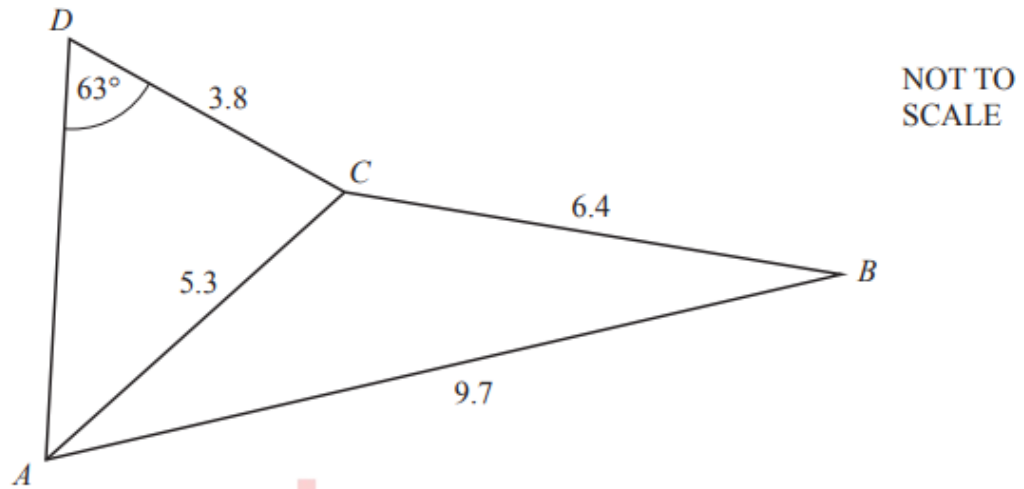
5. S19/qp22/q6(e)



The diagram shows the positions of the boats, A and B, and a harbour, H. $AB = 4\text{km}$, $AH = 6\text{km}$ and $\angle ABH = 70^\circ$.

- Calculate $\angle AHB$. [3]
- The boat at A travels in a straight line to the harbour at H. The average speed of the boat is $p\text{ km/h}$. It takes 12 minutes 20 seconds for the boat to travel from A to H. Calculate p . [3]

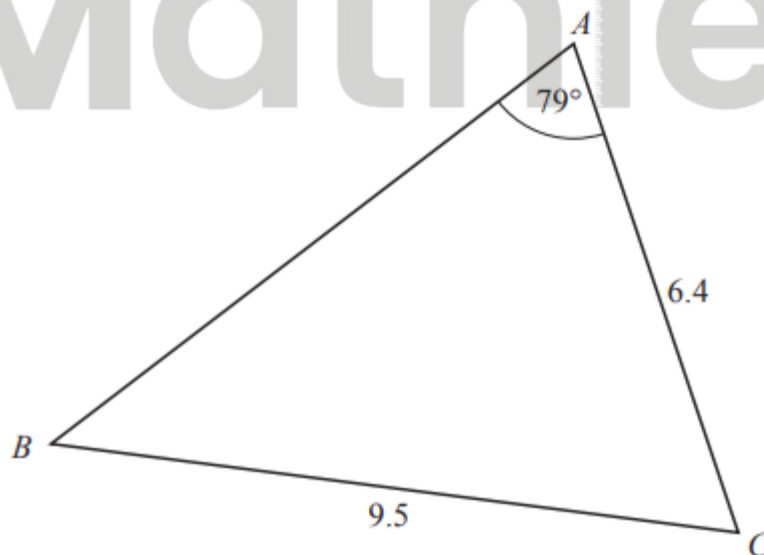
6. W20/qp21/q3(b)



The diagram shows two triangles. $AB = 9.7$ cm, $BC = 6.4$ cm, $CD = 3.8$ cm and $AC = 5.3$ cm. $\angle ADC = 63^\circ$.

- Calculate $\angle DAC$. [3]
- Calculate $\angle ABC$. [3]

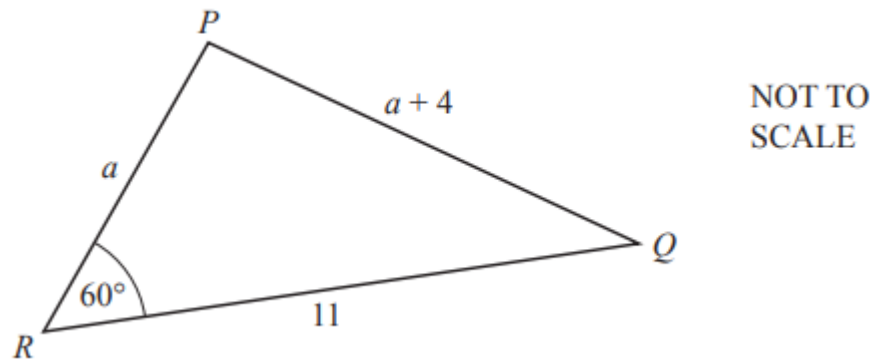
7. W20/qp22/q9(a)



In triangle ABC , $AC = 6.4$ cm, $BC = 9.5$ cm and $\angle BAC = 79^\circ$.

- Calculate $\angle ABC$. [3]
- Calculate the area of triangle ABC . [3]

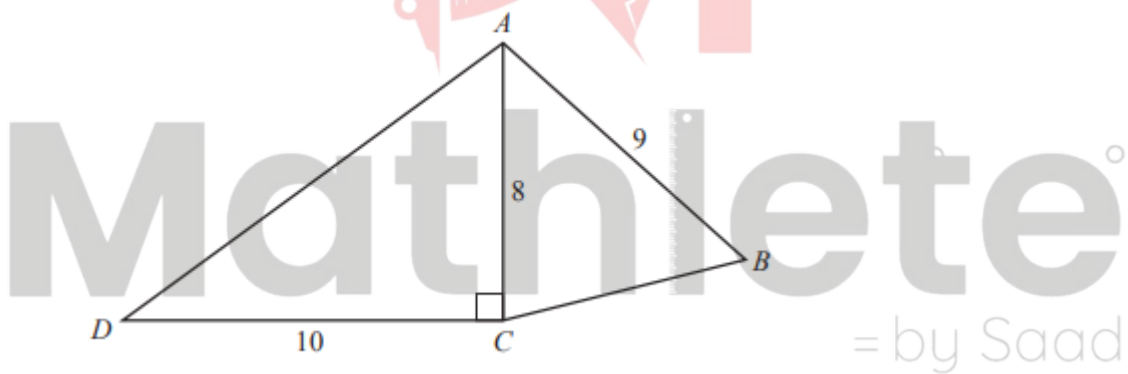
8. S21/qp22/q7(b)



The diagram shows a sketch of triangle PQR . All lengths are given in centimetres. Calculate the length a . [4]

9. S21/qp22/q7(c)

The diagram shows a sketch of quadrilateral $ABCD$. All lengths are given in centimetres.



The area of quadrilateral $ABCD$ is 70cm^2 . Calculate $\hat{DAB}$. [6]

Answer Key

1. S11/qp21/q7

7	(a) 30.4 to 30.45	4	M1 for $16^2 + 20^2 \pm (2) \times 16 \times 20 \cos 115^\circ$ M1 for $\sqrt{656 - 640 \cos 115}$ A1 for 926.(47....)
	(b) $16 \cos 25^\circ$ oe	2	M1 for $\cos 25 = \frac{x}{16}$
	(c) (i) 28 www	2	M1 for $\frac{1}{2}(20 + AD) \times 14.5 = 348$
	(ii) $\frac{1}{2} \times 28 \times 14.5 (= 203)$ or $348 - \frac{1}{2} 20 \times 16 \sin 115$	1	$\frac{1}{2} 30.4 \times 28 \sin 28.5$
	(iii) 28.4 to 28.5	3ft	M1 for $\frac{1}{2} \times 30.4 \times 28 \times \sin CAD = 203$ M1 for $\sin CAD = \frac{203}{\frac{1}{2} \times 30.4 \times 28}$ ft their AC and their AD

2. W11/qp21/q9(a)

9	(a) 122 working seen www	4	M1 for $\frac{\sin ABC}{11} = \frac{\sin 25}{5.5}$ and further M1 for $\sin ABC = \frac{11 \sin 25}{5.5}$ soi and A1 for 58 or B1 for 180 – their 58
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3. S17/qp22/q8

8(a)	14.96 to 15[.0] nfw	3	M2 for $15.1^2 - 2^2 (= 224.01)$ or M1 for $DC^2 + 2^2 = 15.1^2$ or $15.1^2 - \text{their } 2^2$ with horizontal line seen or B1 for horizontal line and 2 soi
8(b)	97.46 to 97.55	3	M2 for $\cos [A] = \frac{9^2 + 11^2 - 15.1^2}{2 \times 9 \times 11}$ oe or B1 for $15.1^2 = 9^2 + 11^2 - 2 \times 9 \times 11 \times \cos [A]$ oe
8(c)	123.8 to 124.1 nfw	4	M3 for $\frac{1}{2} \times 9 \times 11 \times \sin(b) + \frac{1}{2} \times (4 + 6) \times (a)$ oe with (a) $\neq 15.1$ soi or M1 for $\frac{1}{2} \times 9 \times 11 \times \sin(b)$ oe soi and M1 for $\frac{1}{2} \times (4 + 6) \times (a)$ oe with (a) $\neq 15.1$ soi

4. S19/qp21/q5(b)

5(b)(i)	7.95 or 7.948 to 7.949	2	M1 for $\cos 58 = \frac{PS}{15}$ oe
5(b)(ii)	133.7 or 133.72 to 133.73	4	B1 for 32 and M2 for $\sin R = \frac{15 \times \sin their 32}{11}$ or M1 for $\frac{15}{\sin R} = \frac{11}{\sin their 32}$ oe

5. S19/qp22/q6(e)

6(e)(i)	38.8 or 38.78 to 38.79	3	M2 for $\sin[...]=\frac{4\sin 70}{6}$ or $\sin^{-1}\left(\frac{4\sin 70}{6}\right)$ or M1 for $\frac{4}{\sin[...]} = \frac{6}{\sin 70}$ oe
6(e)(ii)	29.2 or 29.18 to 29.19	3	M2 for $\frac{6}{12\frac{1}{3}} \times 60$ oe or M1 for $\frac{6}{their\ time} = \frac{6000}{their\ time}$

6. W20/qp21/q3(b)

3(b)(i)	39.7 or 39.70...	3	M2 for $[\sin =] \frac{3.8 \times \sin 63}{5.3}$ or M1 for $\frac{3.8}{\sin[...]} = \frac{5.3}{\sin 63}$ oe
3(b)(ii)	30.5 or 30.51 to 30.52	3	M2 for $[\cos =] \frac{6.4^2 + 9.7^2 - 5.3^2}{2 \times 6.4 \times 9.7}$ or M1 for $5.3^2 = 6.4^2 + 9.7^2 - 2 \times 6.4 \times 9.7 \times \cos[...]$

7. W20/qp22/q9(a)

9(a)(i)	41.4[0] or 41.39...	3	M2 for $[\sin =] \frac{6.4 \times \sin 79}{9.5}$ or M1 for $\frac{6.4}{\sin[\dots]} = \frac{9.5}{\sin 79}$ oe
9(a)(ii)	26.2[2...]	3	Dep on 3 marks in (a)(i) or 41.4[0] used M1FT for $[\hat{C} =] 180 - 79 - \text{their } 41.4$ M1 for $\frac{1}{2} \times 6.4 \times 9.5 \times \sin(\text{their } \hat{C})$ oe

8. S21/qp22/q7(b)

7(b)	$5 \frac{10}{19}$ or 5.53 or 5.526... nfw	4	M1 for $(a+4)^2 = a^2 + 11^2 - 2 \times a \times 11 \cos 60$ oe B1 for $a^2 + 4a + 4a + 16$ oe M1 for correct isolation of <i>their</i> terms in a to form a linear equation with use of $\cos 60 = 0.5$ After 0 scored, SC1 for $a + 4 = a^2 + 121 - 11a$ oe
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9. S21/qp22/q7(c)

7(c)	107.8 or 107.78 to 107.79...	6	M2 for $\tan^{-1}\left(\frac{10}{8}\right)$ oe or M1 for $\tan \dots = \frac{10}{8}$ oe B1 for $\frac{1}{2} \times 10 \times 8$ soi M2 for $\frac{1}{2} \times 8 \times 9 \sin x = 70 - \text{their } 40$ oe or M1 for $\frac{1}{2} \times 8 \times 9 \sin x$
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